
Report

1.1 K-means clustering and Gaussian Mixture Models

In this part, we built a classifier based on Gaussian mixture models (with K-means used as a preprocessing tool) for synthetic as well as image data. We considered the features of the images to be independent for building our model in order to increase the number of data points. We analyzed our model for different number of mixtures and different proportion of runs for k-means and EM algorithm.

1.1.1 Synthetic Dataset

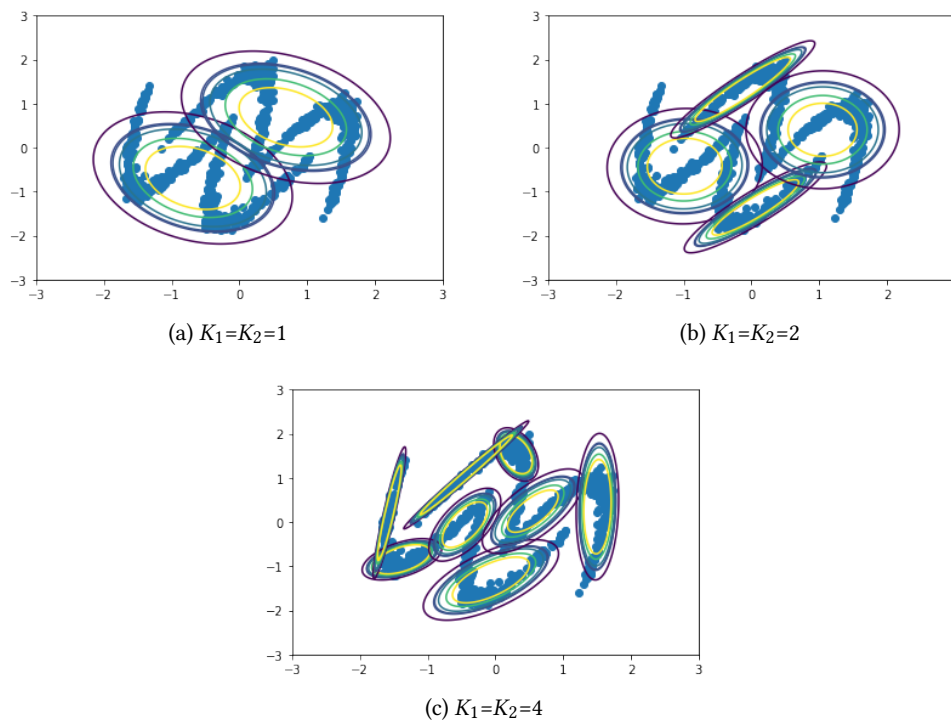


FIGURE 1.1: Sample plots of GMMs for different numbers of mixtures

- Note that K_1, K_2 are the numbers of mixtures of class 1 and 2 respectively
- From Figure 1.1, we infer that 4 mixtures very well fit the data of both the classes.
- From Figure 1.2 and 1.3, we infer that non-diagonal covariances perform slightly better than diagonal covariances

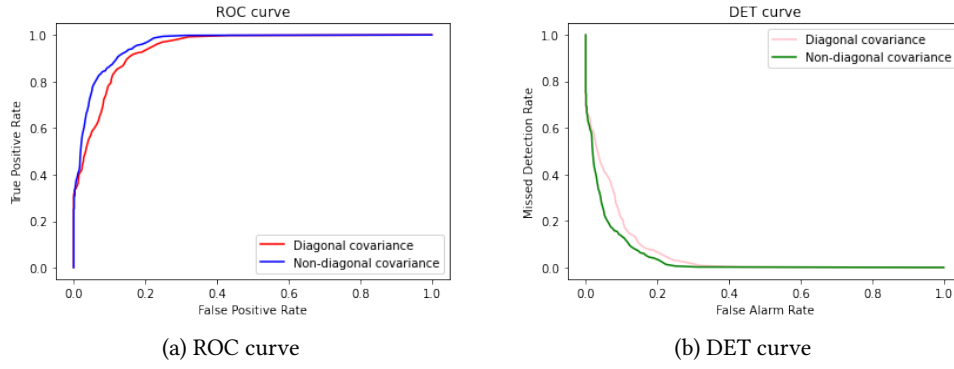


FIGURE 1.2: ROC and DET curves for $K_1=K_2=1$

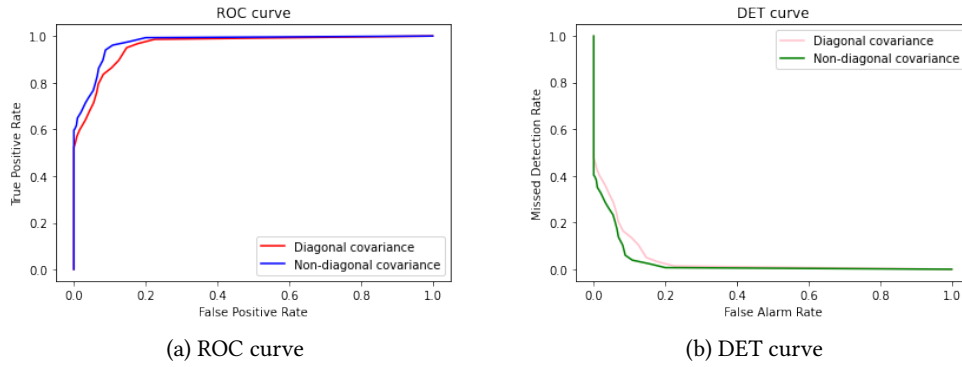


FIGURE 1.3: ROC and DET curves for $K_1=K_2=3$

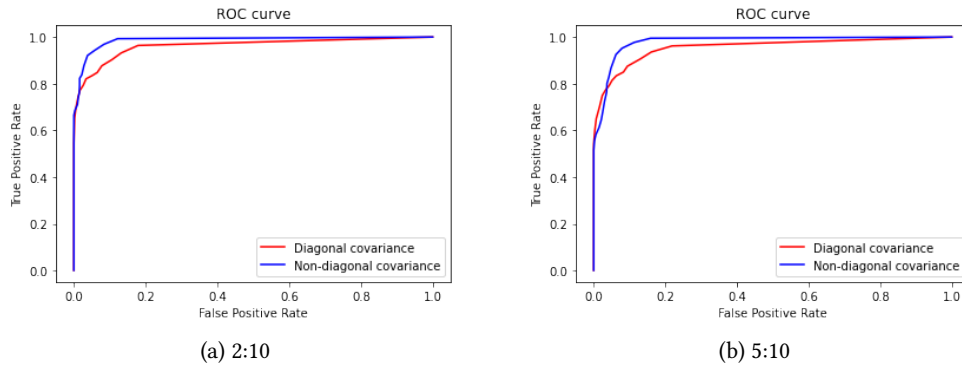


FIGURE 1.4: Different proportion of runs for K-means and GMM algorithm ($K_1=K_2=4$)

From Figure 1.2, we observe that preprocessing by k-means doesn't have a significant effect on the model performance for synthetic data.

1.1.2 Image Dataset

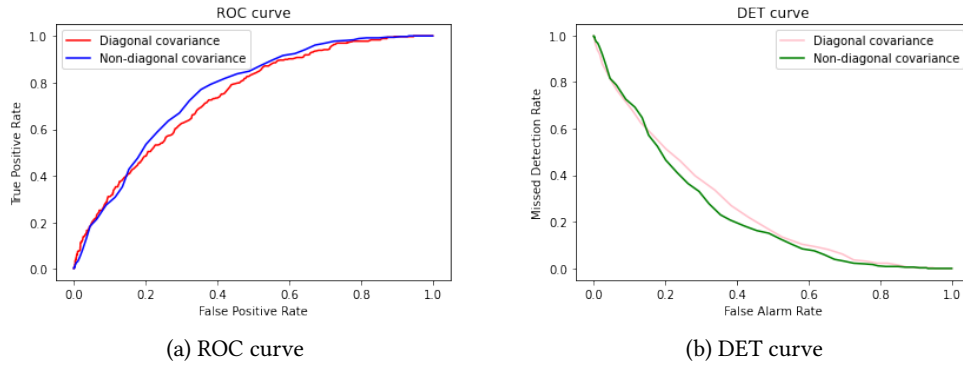


FIGURE 1.5: ROC and DET curves for $K_1=K_2=K_3=2$, $K_4=3$, $K_5=4$

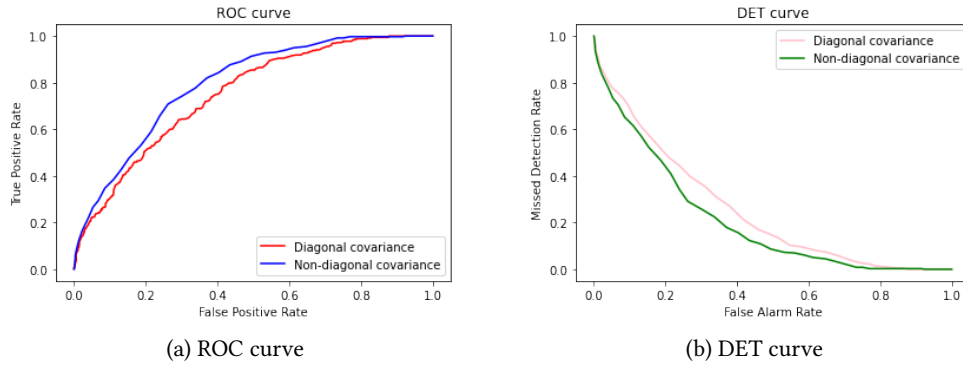


FIGURE 1.6: ROC and DET curves for $K_1=K_2=K_3=K_4=K_5=4$

- The overall performance of the model decreases as compared to the synthetic data
- Not much difference in performance when number of mixtures are varied
- Non-diagonal covariances again perform slightly better than diagonal covariances

1.1.3 Universal Background Model - Gaussian Mixture Model

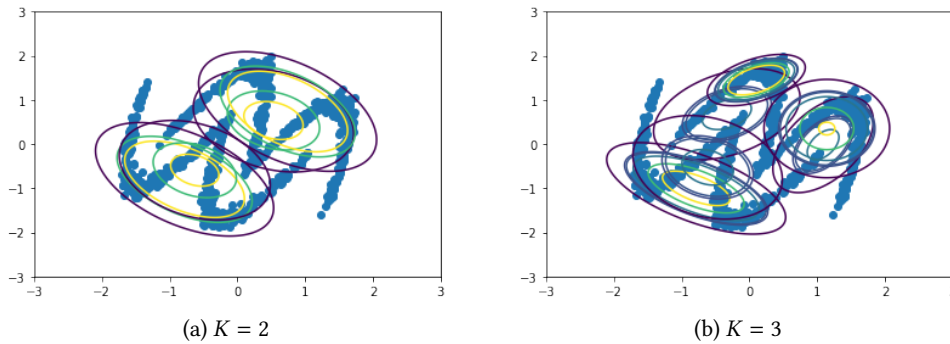


FIGURE 1.7: Sample plots of adapted models for class 1 and class2 for different numbers of mixtures

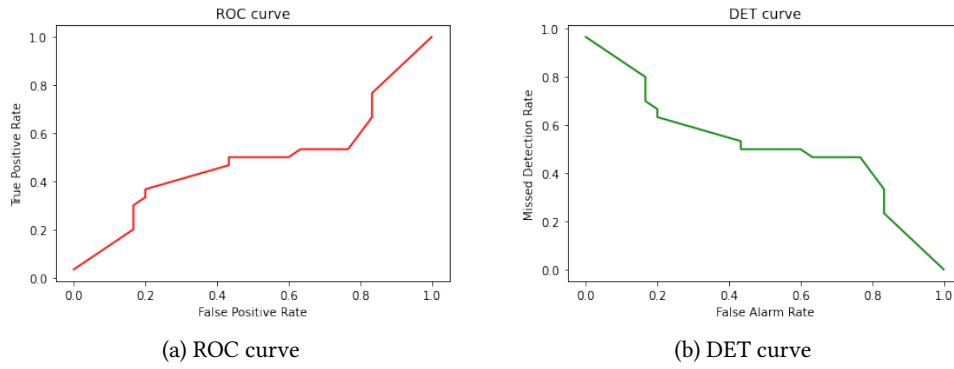


FIGURE 1.8: ROC and DET curves ($K = 2$)

- Usually performs well when trained with a large amount of data belonging to the domain
- Took a large amount of time so had to sample a small subset of data points for score calculation.

We observed a considerable difference in performance between diagonal and non-diagonal covariances. Also the time taken to compute the scores was quite huge because of the nature of the process so, we randomly sampled a small subset of data points and calculated scores for them and then performed analysis (ROC, DET curves).

1.2 Dynamic Time Warping and Hidden Markov Models

We built a classifier of sequential data based on Dynamic Time Warping and Hidden Markov Models. The scores are essentially the frequencies of the particular class (divided by number of top values we are considering) for each test point.

1.2.1 Dynamic Time Warping

1.2.1.1 Isolated Spoken-Digit data

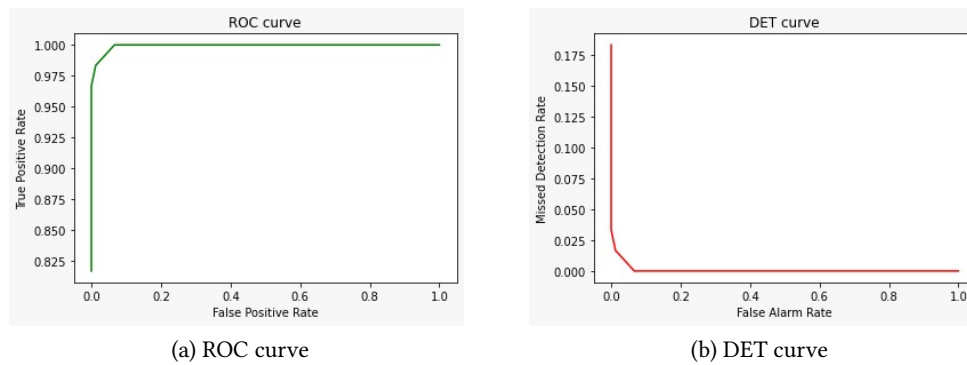
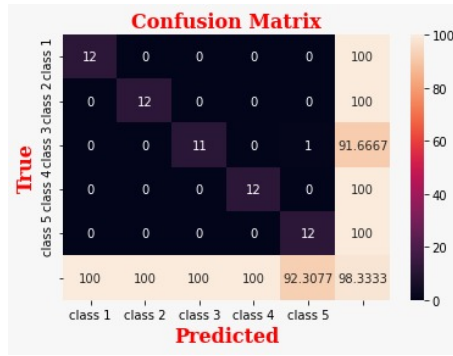


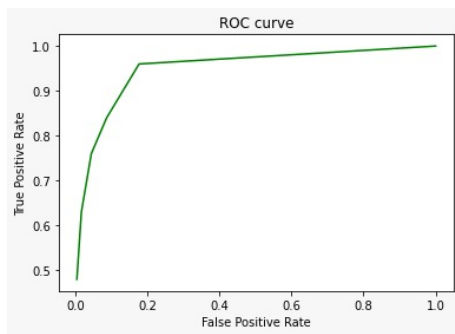
FIGURE 1.9: ROC and DET curve for K (number of top distances to use in classification)=5



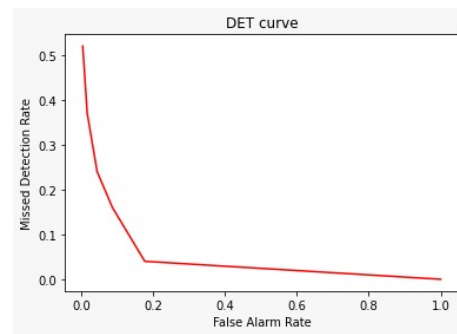
(a) Confusion matrix

FIGURE 1.10: Confusion matrix for K(number of top distances to use in classification)=5

1.2.1.2 Online Handwritten-Character data

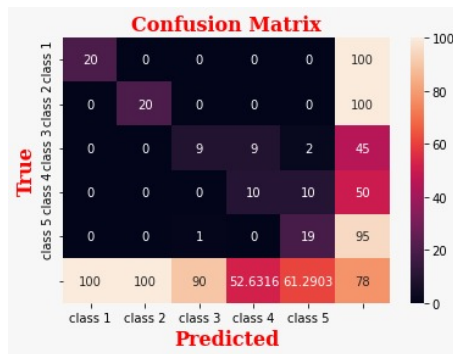


(a) ROC curve



(b) DET curve

FIGURE 1.11: ROC and DET curve for K(number of top distances to use in classification)=5



(a) Confusion matrix

FIGURE 1.12: Confusion matrix for K(number of top distances to use in classification)=5

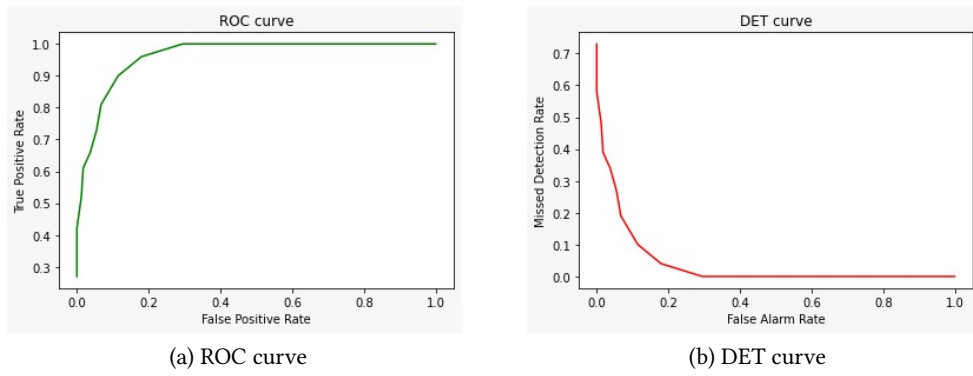


FIGURE 1.13: ROC and DET curve for K(number of top distances to use in classification)=10

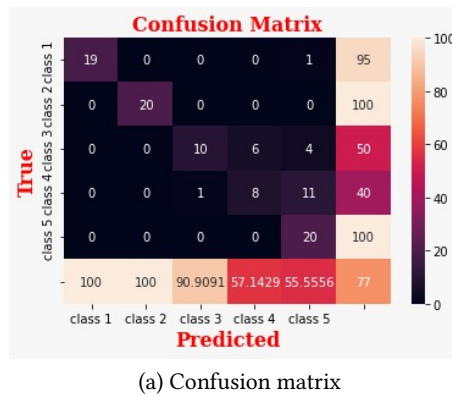


FIGURE 1.14: Confusion matrix for K(number of top distances to use in classification)=10

1.2.2 HMM

1.2.2.1 Method

We produced codebook using K-means clustering for Training Data. Vector quantised both training and dev data using the codebook. Then trained a HMM model for each class separately with parameters Hidden states, Symbols (number of cluster). Tested the dev data with each class trained models and stored the scores.

Samples = 15 for isolated and 10 for Handwritten (using trial and error)

1.2.2.2 Isolated Digits

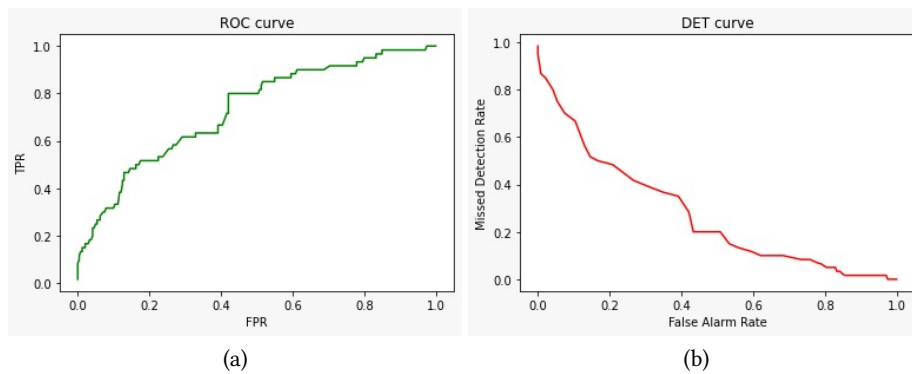
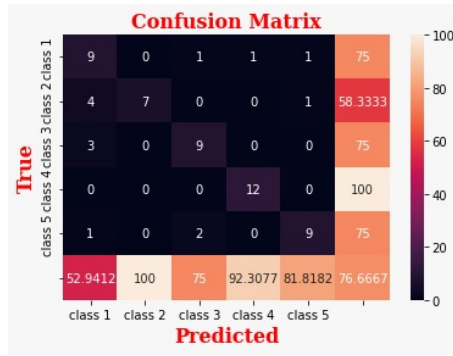
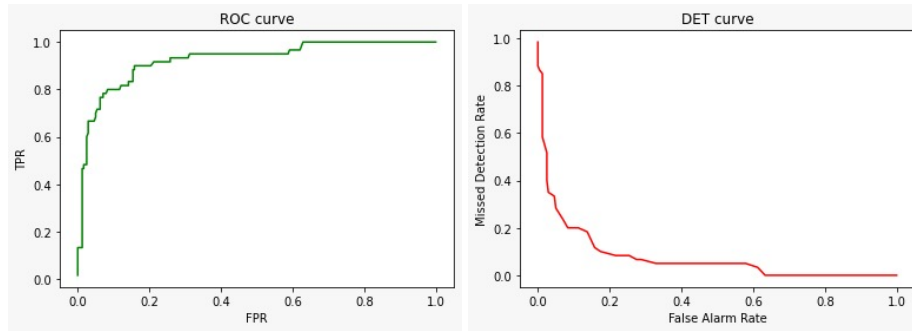


FIGURE 1.15: ROC and DET curve For K (clusters) = 3



(a)

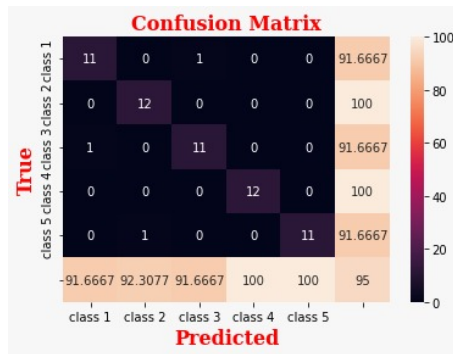
FIGURE 1.16: Confusion Matrix For K (clusters) = 3



(a)

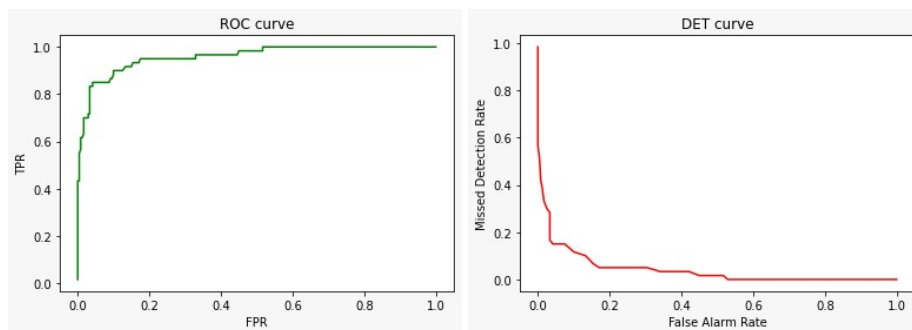
(b)

FIGURE 1.17: ROC and DET curve For K (clusters) = 10



(a)

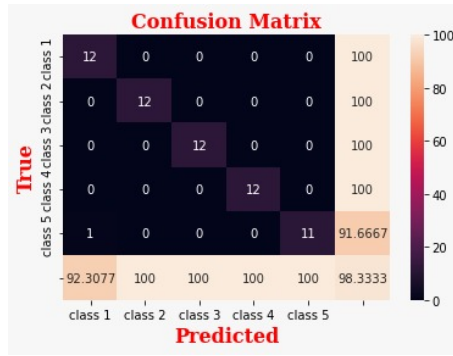
FIGURE 1.18: Confusion Matrix For K (clusters) = 10



(a)

(b)

FIGURE 1.19: ROC and DET Curve For K (clusters) = 20



(a)

FIGURE 1.20: We can clearly see as number of cluster increases from 5 to 20 the efficiency of the model increases

1.2.2.3 Handwritten data

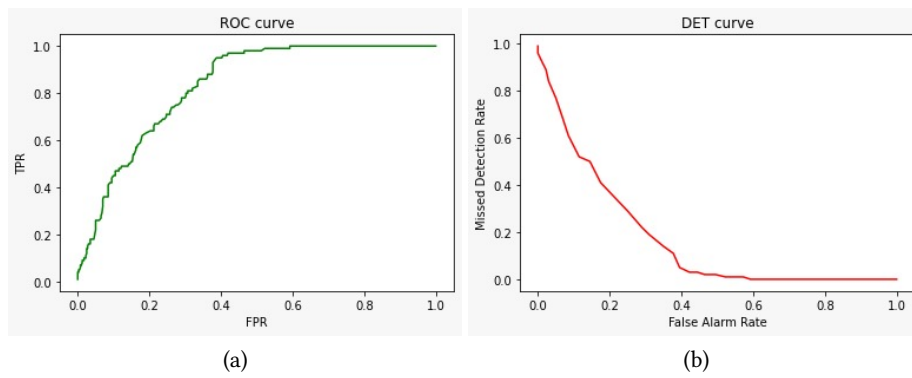
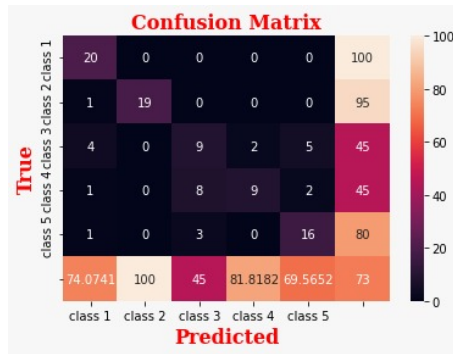


FIGURE 1.21: ROC and DET curve for $k = 20$, without extracting any features



(a)

FIGURE 1.22: Confusion Matrix for $k = 20$, without extracting any features

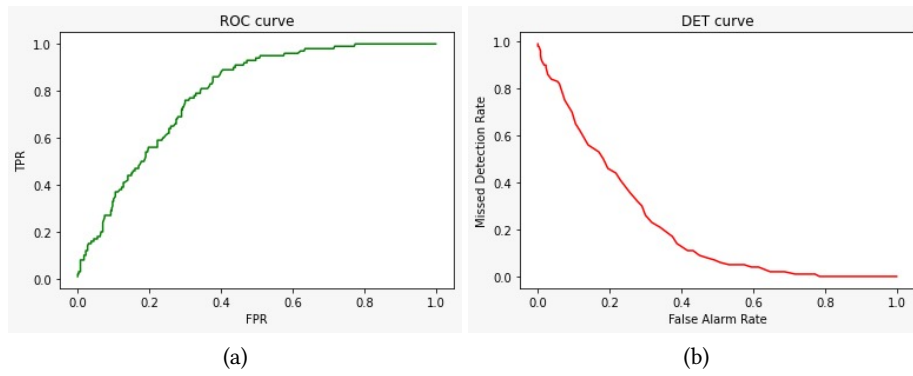


FIGURE 1.23: ROC and DET curve for $k=5$, with slope as extra feature

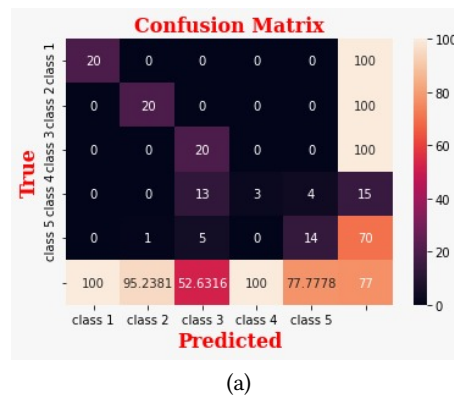


FIGURE 1.24: Confusion Matrix for $k=5$, with slope as extra feature

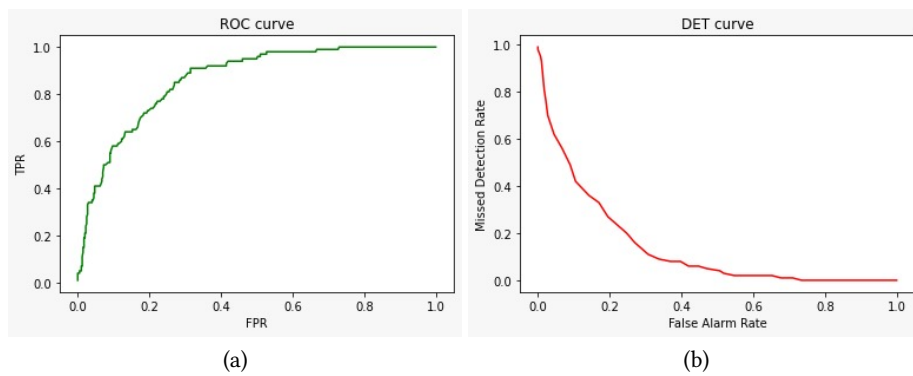


FIGURE 1.25: ROC and DET curve for $k = 20$, with slope as extra feature

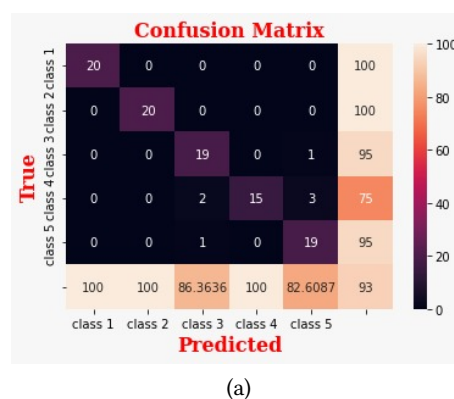


FIGURE 1.26: Confusion Matrix for $k = 20$, with slope as extra feature, we can clearly see extracting more features improves the performance of the model

1.2.3 HMM Bonus

1.2.3.1 Method

As in dev data only two connected or three connected digits/characters are given. I appended the trained HMMs for all possible 2,3 connected sequence, changed samples accordingly and Predicted the sequence for test data.

1.2.3.2 Connected digits

Test Data	prediction 1	prediction 2	prediction 3	prediction 4	prediction 5	prediction 6
1	4 4	4 4 4	o 4	o 4 4	4 o	o o 4
2	7 7	7 3 7	3 7	7 7 7	7 5	7 4 7
3	7 7	7 7 3	7 o	7 7 4	7 5	7 7 5
4	7 7	7 7 7	7 3	7 7 3	7 4	7 7 4
5	5 5	4 5 5	5 3	4 5 3	5 o	4 5 o

1.2.3.3 Connected handwritten data

Test Data	prediction 1	prediction 2	prediction 3	prediction 4	prediction 5	prediction 6
1	a ba a	a ba ai	a ba ba	a ba cha	a ba da	ba ba a
2	ai ba ba	ai ba cha	ai cha a	ai cha da	ai cha ba	ai da a
3	ba cha ba	ba cha da	cha cha ba	cha cha ai	a cha ba	a cha ai
4	ba ba a	ba ba ai	ba da a	ba da ba	ba a a	ba a ai
5	ba ai a	ba ai ba	ba ai cha	ba ba a	ba ba cha	ba ai da